

ZYNTAX Team • Rathinam Technical Campus

PROOFLY

Evidence-First Recruitment Intelligence

"Don't just rank candidates. Prove why."

The Challenge: Black Box Recruitment AI

Traditional Manual Screening

- Extremely time-consuming when evaluating hundreds of resumes against complex job descriptions.
- Prone to human fatigue, oversight, and subjective bias during initial filtering rounds.

Conventional AI Screening

- Operates like an opaque "black box" yielding scores without explaining reasoning (e.g., "87% Match").
- Recruiters are forced to manually verify every single claim and lack traceability back to resume text.

Revolutionary Approach

The PROOFLY Proposed Solution

AI assists → Evidence proves → Humans decide

PROOFLY establishes a transparent, end-to-end evidence chain linking match scores directly to verbatim resume clauses.

System Architecture



Frontend Layer

React + Tailwind CSS

Provides intuitive dashboards for Candidates, Recruiters, and Admins to manage profiles, jobs, and reports.



Backend & API

Node.js + Express

Handles secure REST APIs, authentication via JWT, file parsing, and core orchestration logic.



AI & Data Engine

NLP Pipeline & PostgreSQL

Extracts structured resume sections, maps evidence chains, and calculates the PROOF SCORE™.

Core Feature: "Prove This Decision"

Traceable Evidence Chain Flow

Overall Score → Job Requirement → Candidate Field → Resume Section → Exact Snippet

Eliminates black-box speculation by granting recruiters verifiable proof to defend every shortlisting decision instantly.

Advanced Skill Gap Analysis



Strong Match

Python, SQL, REST API

Directly supported by verified evidence and project citations in the candidate's resume.



Partial Match

AWS / Cloud Services

Mentioned briefly in passing or lacking complete architectural project depth.



No Evidence

Kubernetes, Terraform

Not identified in submitted documentation (absence of evidence is not proof of absence).

Platform User Roles



Candidate

Register, upload PDF/DOCX resumes, apply for open positions, and track application evaluation statuses.



Recruiter

Create jobs, define specific technical requirements, execute evaluations, compare candidates, and make final decisions.



Admin

Manage platform users, monitor parser and processing health, review failed documents, and oversee system audits.

Key Functional Requirements

- ✓ **Authentication & Roles:** Secure user registration and login supporting role boundaries.
- ✓ **Resume Processing:** Automated PDF/DOCX text extraction and section parsing.
- ✓ **Job Management:** Recruiters can define required and preferred technical criteria.
- ✓ **Evaluation Engine:** Automated matching algorithms producing the PROOF SCORE™.
- ✓ **Evidence Linking:** Direct storage of extracted evidence associated with resume claims.
- ✓ **Reporting:** Generation of explainable candidate evaluation reports.

Robust Technology Stack

Layer	Selected Technology
Frontend	React + Tailwind CSS
Backend	Node.js + Express REST API
Database	PostgreSQL
Security	JWT Tokens & Secure Password Hashing
Document Processing	PDF & DOCX Parsers
Version Control	Git & GitHub

Security & Quality Assurance

Security & Privacy

- Role-based access control protecting confidential recruiter data from candidates.
- Secure password hashing and strict isolation of API secrets and environment keys.

Testing Plan

- Comprehensive test cases covering authentication, resume extraction accuracy, and scoring logic.
- Validation checks for invalid formats, authorization boundaries, and report generation.

Advantages & Future Roadmap

Key Advantages

- **Explainable:** Complete transparency into AI inference.
- **Evidence-Driven:** Direct ties to original resume text.
- **Recruiter Controlled:** AI assists rather than automates final hires.

Future Enhancements

- Advanced Skill Graphs & Requirement DNA analysis.
- Proof Heatmaps & Interactive Job Gap Simulators.
- Large-scale enterprise ATS integrations.

Conclusion

"The score is only the beginning. The proof is the product."

Questions & Discussion

Thank you! — Team ZYNTAX (Sheshagiri, Yokesh, Subasri)